MIL-F-25562B(USAF)
27 February 1964
Superseding
MIL-F-25562A(USAF)
21 January 1958

MILITARY SPECIFICATION

FILTER, RADIO INTERFERENCE F-245/APX-25

1. SCOPE

1.1 This specification covers the requirements for a filter unit, designated Filter, Radio Interference F-245/APX-25.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-F-15733	Filters, Radio Interference, General Specification for
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-R-26667	Reliability and Longevity Requirements, Electronic Equipment, General Specification for
MIL-R-27542	Reliability Program Requirements for Aerospace Systems, Subsystems and Equipment

STANDARDS

Military

MIL-STD-220	Method of Insertion - Loss Management for Radio-Frequency Filters
MIL-STD-831	Test Reports, Preparation of
MS3108	Connector, Plug, Electric, Angle 90°

FSC 5915

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DRAWING

Air Force

55C12624 Installation Information, Filter, Radio Interference F-245/APX-25

(Copies of documents required by contractors in connection with specific functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 PREPRODUCTION.- This specification makes provision for preproduction testing.

3.2 GENERAL SPECIFICATIONS.- Filter, Radio Interference F-245/APX-25 shall be in accordance with MIL-E-5400 and MIL-F-15733 with the exceptions and additions called out herein. In case of conflict, this specification shall govern.

3.3 DESIGN.- Filter, Radio Interference F-245/APX-25 shall be designed for insertion in the power leads to Transponder Set AN/APX-25 to provide reduction of r-f interference both to and from the transponder set. The design, including construction and selection of materials, shall provide reliability and longevity with freedom from maintenance within the requirements herein. The a-c and d-c sections of the filter shall have the following characteristics:

A-c filter:	
Voltage	102-124 volts
Current	4 amps
Frequency	320-1760 cps
D-c filter:	
Voltage	100 volts
Current	4 amps

3.3.1 ISOLATION.- Isolation between the a-c and d-c filter sections shall be sufficient to reduce all r-f cross-coupling between these filter sections by an attenuation factor of at least 50 db over the frequency range of 0.3 to 20 megacycles.

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3.3.2 INSULATION RESISTANCE.- The insulation resistance between the filter elements and between each filter element and the case shall be at least 3000 megohms.

3.3.3 VOLTAGE DROP.- The maximum voltage drop through the d-c filter shall not exceed 0.2 volt and the maximum voltage drop through the a-c filter shall not exceed 1.0 volt at any frequency within the range specified in 3.3.

3.3.4 INSERTION LOSS.- The insertion loss shall be not less than 50 db over the frequency range of 0.3 to 20 megacycles and not less than 65 db at 0.7 megacycle when tested in accordance with MIL-STD-220 for both the a-c and d-c sections.

3.3.5 TEMPERATURE RISE.- The temperature rise shall not exceed 15°C above the ambient when measured in accordance with MIL-F-15733.

3.3.6 OPERATING TEMPERATURE RANGE.- The filter shall provide specified performance throughout the ambient temperature range of A (-55 to +85°C) in the table entitled Operating Temperature Range of MIL-F-15733.

3.3.7 GRADE.- The filter shall conform to grade 2 in the table entitled Grade of MIL-F-15733, 10 to 500 cps.

3.4 CONSTRUCTION

3.4.1 CASE.- The filter shall be constructed in accordance with Drawing 55C12624 for the maximum overall dimensions and mounting provisions. The case shall provide a complete electrical and mechanical enclosure and shall be made of corrosion-resistant material or suitably treated to prevent corrosion. An electrical receptacle shall be mounted on one end of the filter case and a fitting for entrance of the shielded cable shall be provided on the other end. The case shall be filled with an impregnating or potting compound and sealed. Except for air space due to shrinkage from cooling, the potting compound shall contain no significant voids or air pockets.

3.4.2 ELECTRICAL CONNECTIONS.- A receptacle, mounted on one end of the filter case, shall mate with an MS3108B-14S-5S plug without mechanical interference. The other end of the filter case shall connect to an 18-inch shielded cable that terminates in an MS3108B-14S-5S plug. The cable on one end and the shell of the MS3108B-14S-5S plug on the other. The cable and plug shall meet all of the environmental conditions along with the filter.

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3.4.3 WIRING.- Wiring of the filter shall be as follows:

MS3108B-14S-5S		Receptacle mating with MS3108B-14S-5S
Pin A	Through a-c filter	Pin A
Pin B	No connection	Pin B
Pin C	Ground	Pin C
Pin D	Through d-c filter	Pin D
Pin E	No connection	Pin E

3.5 IDENTIFICATION.- The filter shall be identified in accordance with MIL-N-18307. The identification marking may be silk screened, etched, or die struck directly into the metal case and filled with a contrasting color. The method that is used will be subject to the approval of the procuring activity and shall be permanent, legible, of contrasting color to the background, resistant to wear and impervious to the action of water, salt spray, and solvents, usually encountered, such as gasoline, alcohol and lacquer thinners. The following information shall be incorporated:

Filter, Radio Interference F-245/APX-25
Stock No. 3330-058750293
(Manufacturer's Name)
(Manufacturer's Part No.)
Contract No.
Rating, DC 100V, 4 amps
AC 124V, 4 amps, 320 to 1760 cps

3.6 RELIABILITY PROGRAM.- The contractor shall establish a reliability assurance program and submit reports thereon in accordance with MIL-R-27542.

3.6.1 RELIABILITY AND LONGEVITY.- Filter, Radio Interference F-245/APX-25 is intended to be a nonrepairable item. Therefore, the first failure represents the end of life (longevity) for the filter. The filter shall be constructed, by means of its design and selection of parts and materials, to have a longevity and mean-time-to-failure (MTTF) of not less than 5000 hours of operating life.

3.7 MAINTAINABILITY.- For failure other than the MS3108B-14S-5S, which is replaceable, the filter shall be a throw-away item of equipment.

3.8 WEIGHT.- The filter shall not exceed 1 pound in weight.

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4. QUALITY ASSURANCE PROVISIONS

4.1 RESPONSIBILITY FOR INSPECTION.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 GENERAL SPECIFICATION.- Inspection and testing shall be in accordance with MIL-F-15733 with the exceptions and additions herein except that for the purposes of this specification the qualification tests of MIL-F-15733 will constitute preproduction tests in this specification.

4.3 TEST PROGRAM.- Prior to any formal testing of the filters, the contractor shall submit a draft of his proposed preproduction and acceptance test program to the procuring activity for approval. The test program shall include the detailed procedure, specify the equipment to be used in the tests, include diagrams as necessary, and samples of forms for test data. The test program shall be in two parts, the first delineating preproduction and the second delineating acceptance testing.

4.4 CLASSIFICATION OF TESTS.- The inspection and testing of filters shall be classified as follows:

a. Preproduction tests

b. Acceptance tests.

4.5 TEST CONDITIONS.- The following test conditions shall apply:

a. Connector MS3108B-14S-5S shall be considered the input for making insertion loss measurements

b. In conducting the seal test, an atmospheric pressure equivalent to 60,000 feet of altitude shall apply

c. The salt spray test shall be conducted for 48 hours.

4.6 PREPRODUCTION TESTING

4.6.1 TEST SAMPLE.- The test sample shall consist of a minimum of three filters representative of the production equipment. The samples shall be identified with the manufacturer's part number and such other information as required by the procuring activity.

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to perform any of the inspections set forth in the specification where such in-

4.6.2 TEST REPORT AND TEST SAMPLE FOR THE PROCURING ACTIVITY.- When the tests are conducted at a location other than the laboratory of the procuring activity, the following shall be furnished to that activity:

a. Test report: Three copies, one of which shall be a reproducible, of a test report in accordance with MIL-STD-831

b. Test sample: The sample that was tested.

4.6.3 PREPRODUCTION TESTS.- The preproduction tests shall consist of the tests of group I of the table entitled Qualification Inspection of MIL-F-15733, in the order shown, on all 3 sample filters. Then, 1 sample filter shall be subjected to the tests outlined in group II, the second sample filter to the tests of group III, and the third sample filter to the tests of group IV, in the order specified. If the contractor finds it convenient to submit additional sample filters, this is permissible.

4.7 ACCEPTANCE TESTS.- Acceptance tests shall consist of:

a. Individual tests

b. Sampling plan and tests.

4.7.1 INDIVIDUAL TESTS.- Each filter shall be subjected to the following tests:

4.7.1.1 EXAMINATION OF PRODUCT.- Each filter shall be thoroughly examined to determine compliance with the requirements for materials, workmanship, dimensions, wiring, and markings, using whatever tools, gauges, scales, and the like that are necessary.

4.7.1.2 INSERTION LOSS.- Each filter shall be tested for insertion loss in accordance with MIL-F-15733.

4.7.2 SAMPLING PLAN AND TESTS

4.7.2.1 SAMPLING PLAN.- At least 1 filter from each production lot of 200 filters, after the first 200 filters have been produced, shall be subjected to the following tests:

4.7.2.2 SAMPLING TESTS

a. Seal

b. Temperature rise

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c. Dielectric withstanding voltage

d. Insulation resistance

e. Voltage drop

f. Overload

g. In addition to the test specified in 4.7.1.2, an insertion loss measurement shall be made at 0.7 mc/s.

4.8 RELIABILITY AND LONGEVITY TEST PROGRAM.- At least 30 days prior to the scheduled date of testing or at such later date as the contracting officer may authorize, the contractor shall submit a detailed reliability test program to the procuring activity for approval. Testing shall begin only after formal approval of this test program by the procuring activity.

4.9 RELIABILITY TESTS.- Due to the passive nature of Filter, Radio Interference F-245/APX-25 and the excessive time required to effectively demonstrate the MTTF of the filter, preproduction reliability tests of the filter will not be required. However, at least two of the first production samples of the filter shall be placed on a longevity test to be run concurrently with preproduction and production testing. Longevity testing and acceptance reliability testing shall be conducted in accordance with MIL-R-26667 except as specified herein.

4.9.1 EQUIPMENT SELECTION AND TIME FOR RELIABILITY TESTING.- All equipments submitted for acceptance shall each be given an operational test at the specified test level for a period of not less than 50 hours. Time accumulated during acceptance testing as specified in 4.7.1, although not at the test level stated in the specification, may be counted as operational for reliability demonstration purposes. The first 30 minutes of operation shall be considered as burn-in. A cumulative log shall be kept of all operating time except that burn-in and testing time shall be kept separately.

4.9.2 TEST LEVEL.- Filters shall be tested at test level III, table 2.3, of MIL-R-26667 with a chamber temperature range of -54° to +55°C.

4.9.3 PERFORMANCE CHARACTERISTICS TO BE MEASURED DURING RELIABILITY TESTING.- The performance characteristics of insertion loss at 0.7 megacycle and insulation resistance shall be measured at least once during each 24-hour period of operation.

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4.9.4 FAILURES.- All failures shall be reported. Failures occuring during burn-in and failures of connector MS3108B-14S-5S, except soldering thereto, shall not be considered as relevant. All other failures shall be considered as relevant.

4.9.5 ACCEPTANCE CRITERIA.- Deliveries may be made on a basis of no relevant failures until 11,500 hours of testing time have accumulated. After 11,500 hours of testing time have accumulated, acceptance shall be judged on the basis of table 5.1 entitled Reliability Accept-Reject Criteria (Method 1) of MIL-R-26667. Recording, data handling, and reporting procedures shall be in accordance with the recommendations of MIL-R-26667. When a failure occurs before 11,500 hours of reliability testing have accumulated, the contractor shall stop further delivery, determine the nature and cause of the failure, and make a report to the procuring activity. The report shall be reviewed by both the procuring activity and the contractor, wherein a decision shall be made either to redesign or modify the equipment or continue production and delivery. Should either unit or longevity test fail before the specified longevity, the above course of action shall also apply. Units which fail during test shall remain the property of the contractor.

5. PREPARATION FOR DELIVERY

5.1 PREPARATION.- Preparation for delivery shall be in accordance with the requirements of MIL-E-17555.

6. NOTES

6.1 INTENDED USE.- This filter is used in the power input leads to a radar beacon to reduce the conducted interference originating in the radar beacon.

6.2 ORDERING DATA.- Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Number of preproduction samples, point of testing, and delivery of preproduction test report (see 4.6.1 and 4.6.2)

c. Submission of test programs for approval (see 3.6, 4.3, and 4.8)

d. Selection of applicable levels of preservation, packaging, packing, and marking.

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SPECIFICATION ANALYSIS SHEET	Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use of the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION		
ORGANIZATION		CITY AND STATE
CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A
☐ Direct Government Contract ☐ Subcontract

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO IF "YES" IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity)	DATE

DD Form 1426

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